

Pseudo Visible Feature Fine-Grained Fusion for Thermal Object Detection

Ting Li¹, Mao Ye^{1*}, Tianwen Wu¹, Nianxin Li¹, Shuaifeng Li¹, Song Tang², and Luping Ji¹

¹ University of Electronic Science and Technology of China

² University of Shanghai for Science and Technology

lting1103@gmail.com, cvlab.uestc@gmail.com

Abstract

Thermal object detection is a critical task in various fields, such as surveillance and autonomous driving. Current state-of-the-art (SOTA) models always leverage a prior Thermal-To-Visible (T2V) translation model to obtain visible spectrum information, followed by a cross-modality aggregation module to fuse information from both modalities. However, this fusion approach does not fully exploit the complementary visible spectrum information beneficial for thermal detection. To address this issue, we propose a novel cross-modal fusion method called Pseudo Visible Feature Fine-Grained Fusion (PFGF). Specifically, a graph is constructed with nodes generated from multi-level thermal features and pseudo-visual latent features produced by the T2V model. Each level of features corresponds to a subgraph. An Inter-Mamba block is proposed to perform cross-modality fusion between nodes at the lowest level; while a Cascade Knowledge Integration (CKI) strategy is used to fuse low-level fused information to high-level subgraphs in a cascade manner. After several iterations of graph node updating, each subgraph outputs an aggregated feature to the detection head respectively. Unlike previous cross-modal fusion methods, our approach explicitly models high-level relationships between cross-modal data, effectively fusing different granularity information. Experimental results demonstrate that our method achieves SOTA detection performance. Code is available at <https://github.com/liting1018/PFGF>.

1. Introduction

Object detection is critical in computer vision, with applications in autonomous driving and video surveillance [27]. While most methods rely on visible spectrum images, they often struggle in adverse conditions like nighttime, fog, or snow. In contrast, thermal spectrum images offer robustness in challenging environments [13], but they may falter in the daytime due to limited texture detail. Both visible-

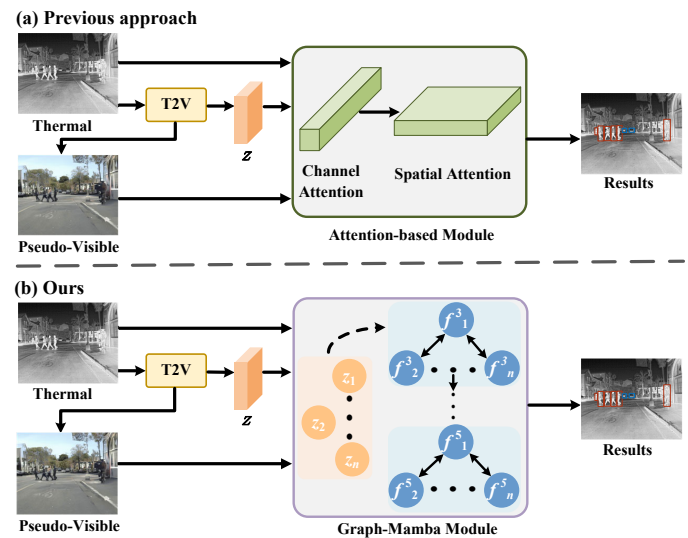


Figure 1. Comparison between previous attention-based method and our method. (a) An attention-based module combines channel and spatial attention to augment the modality-translation awareness of detection. (b) Our method employs a Graph-Mamba Fusion module to integrate cross-modality knowledge in a cascade manner with different granularity.

only and thermal-only approaches face limitations by solely relying on one spectrum, and lacking complementary data. As a result, multi-spectral (visible and thermal) object detection [8, 25, 52] have gained popularity. However, these approaches require paired visible and thermal images for both training and inference, increasing hardware costs and complexity. Furthermore, the need for precise alignment between modalities further limits practical deployment [5]. Consequently, there is growing interest in visible knowledge assistance methods that utilize paired visible-thermal images for training and solely thermal images for testing.

Visible knowledge-assisted thermal pedestrian detection methods can be categorized into two routes: *data enhancement* and *cross-modal transfer*. The route of data enhancement [2] employs generative models to adapt visible images to the thermal domain like domain adaptation [26, 29], us-

*The corresponding author.

ing them during training to mitigate the limitations caused by relying solely on thermal data during testing. However, the generated thermal images may be unreliable, potentially degrading the performance of thermal pedestrian detectors. Another route is based on cross-modal transfer, which can be further divided into two approaches. The first approach, feature domain transformation [45], utilizes an auxiliary network to convert thermal features into visible ones. However, this process demands substantial computation and large models to maintain feature quality in high-dimensional spaces. The second approach, image domain transformation [44], employs a generative model to directly convert thermal images into visible ones. The transformed visible images are then fused with the original thermal images, as illustrated in Fig.1(a).

While image domain transformation methods have attained the state-of-the-art performance in thermal object detection, they still face obstacles related to occlusions and visible image noise. This limitation primarily stems from the inherent unreliability of visible images generated from thermal data, which can introduce noise that adversely affects detection accuracy. Furthermore, these approaches often neglect the fine-grained multimodal knowledge relationships between pseudo-visible and thermal data. As a result, effectively harnessing the complementary nature of thermal and visible information to resolve ambiguities in complex scenes remains a significant challenge. Fortunately, Graph Neural Networks (GNNs) offer a promising alternative for modeling and reasoning about relationships [30]. This motivates us to utilize graph structures for enabling interactions among features at various granularities. Such an approach ensures comprehensive utilization of information between low-level features to high-level semantics, leading to richer and more nuanced cross-modal integration. However, the flexible flow of information between nodes may introduce redundancy or irrelevant data, potentially hindering detection performance. To address this, we incorporate Mamba’s selection mechanism [53] to effectively filter out unimportant information in the long-range context, which is particularly crucial for thermal feature maps that often contain noise and lack texture details. Furthermore, by iteratively updating node features based on neighborhood nodes, it can minimize the impact of less relevant nodes, thereby enhancing feature fusion and representation.

Based on the above analysis, as illustrated in Fig. 1(b), we propose a novel cross-modal fusion method, dubbed as Pseudo Visible Feature Fine-Grained Fusion (PFGF). First, we obtain the visible features using the Pearl-GAN method [37], which are referred to as generative features. To fuse these generative features, we introduce the Graph-Mamba Fusion module. Specially, a graph is constructed that consists of multiple subgraphs, where each subgraph corresponds to either the generative feature or the output features

from different levels of the detector backbone (referred to as discriminative features). The node features in each subgraph represent multi-level information, corresponding to different feature levels. For graph computation, we propose the Inter-Mamba block and Cascade Knowledge Integration (CKI) strategy, which facilitate efficient knowledge fusion and propagation across subgraphs. The Inter-Mamba block is introduced to fuse the nodes of subgraphs between the generative feature and the low-level discriminative feature, here the c^3 feature from CSPDarknet [1]. Then, a Cascade Knowledge Integration (CKI) strategy is proposed to propagate the knowledge from the low-level subgraph to the high-level subgraph in a cascade manner. During this cascading process, node features are updated using the Gate Recurrent Unit (GRU) method [9], while the VMamba block further refines these features across subgraphs.

Our contributions can be summarized as follows: (1) We are the first to employ pseudo visible feature fine-grained fusion method for thermal object detection. The information interaction ability of graph at different granularities and the information selection and enhancement ability of Mamba have been fully utilized. (2) An Inter-Mamba block is proposed to facilitate cross-modality fusion between generative and discriminative node features. Additionally, the cascade knowledge integration strategy aggregates knowledge from low-level to high-level feature subgraphs avoiding tedious computation. (3) Experiments conducted on public FLIR, LLVIP, and Autonomous Vehicles datasets demonstrate significant advantages over the state-of-the-art (SOTA) thermal object detection methods.

2. Related Work

2.1. Thermal Object Detection

Thermal object detection is crucial in low-light environments and privacy-sensitive scenarios. Existing approaches can be categorized into *direct reconstruction* and *visible knowledge assistance*. In the direct reconstruction approach, the methods [21, 22, 38] focus on designing networks to extract discriminative features by leveraging the intrinsic characteristics of thermal images. For example, GPCAnet [48] predicts object positions in the image to allocate more ROIs to the predicted areas. The *Visible knowledge assistance* methods can be further divided into *data augmentation* and *cross-modal transfer*. The data augmentation works [2, 20] use GANs to convert visible images into thermal images, mixing them to enhance detector robustness. In the cross-modal transfer approach, the works try to generate pseudo visual images or features. For example, TIRDet [44] uses Pearl-GAN for image translation. Xie et al [45] use a hallucination network to learn visible features from thermal images. All methods lack in-depth research on pseudo-visible information fusion.

2.2. Graph Based Cross Modality Fusion

In recent years, graph-based approaches have gained significant traction in computer vision applications [36]. Typically, graph operates as message-passing algorithms, where node representations are iteratively updated based on their neighbors through a differentiable aggregation function [34]. This makes graph particularly valuable in multi-source data fusion. For instance, Cai et al. [3] used a graph-attention mechanism to fuse embeddings from different feature extractors, while Li et al. [24] proposed an interactive GNN-based architecture to facilitate cross-modality fusion. Our work is closely related latter work that utilizes graph structures to explicitly model and reason about different granularity information interactions between complementary data sources. However, our approach uniquely incorporates Mamba's selection mechanism to strengthen graph node representations, effectively minimizing the impact of unimportant nodes at each step of hidden state computation.

2.3. Mamba Based Cross Modality Fusion

Since its inception for linear-time sequence modeling in natural language processing, Mamba [14] has rapidly been adapted for various applications in computer vision. For example, VMamba [35] leverages a four-way scanning algorithm to advance object detection and segmentation beyond the capabilities of the Swin Transformer. Mamba has already been used in cross-modality fusion. Xie et al. [46] leveraged Mamba for dynamic feature enhancement in multimodal image fusion. Pan-Mamba [15] is developed to facilitate efficient cross-modal information exchange and fusion. The Fusion-Mamba block [10] is introduced to mitigate disparities between cross-modal features and enhance the consistency of fused feature representations. Additionally, Li et al. [23] proposed cross-modality fusion-Mamba, coupled with weather removal techniques, to improve detection accuracy under adverse weather conditions. Distinct from prior research, our work pioneers the integration of the Mamba architecture into graph representations for thermal object detection tasks, significantly enhancing graph embeddings and processing capabilities.

3. Methodology

3.1. Problem Statement

Suppose the training dataset $D_s = \{X^t, X^v\}$, where the thermal image set $X^t = \{x_{s_i}^t\}_{i=1}^{N_{st}}$ with labels $\{y_{s_i}^t\}_{i=1}^{N_{st}}$ and the unpaired visible image set $X^v = \{x_{s_i}^v\}_{i=1}^{N_{sv}}$. N_{st} and N_{sv} denote the cardinality of the thermal image set and visible image set respectively. The test dataset $D_t = \{x_{t_i}^t\}_{i=1}^{N_t}$ be exclusively composed of thermal images, where N_t represents the number of thermal images in the test dataset. Our objective is to compensate for visible knowledge while using solely thermal images as the input at the test stage,

and fully leverage information from both modalities to develop robust representations at the training stage.

Overview. Fig. 2(a) illustrates the proposed Pseudo Visible Feature Fine-Grained Fusion (PFGF) framework, which uses the single-stage detector YOLOX [12] as the baseline model. Both training and testing are conducted exclusively with thermal images. Specifically, we first train Pearl-GAN as a prior thermal-to-visible (T2V) translation model Φ , which is trained on both thermal and visible modalities, to convert thermal image x^t into daytime visible image x^v , thereby incorporating valuable visible information from the unpaired set X^v . This process is described as follows:

$$[x^v, z] = \Phi(x^t), \quad (1)$$

where $x^t \in \mathbb{R}^{H \times W \times 1}$ and $x^v \in \mathbb{R}^{H \times W \times 3}$. $\Phi(x^t)$ comprises a convolution-based encoder $z = \varepsilon(x^t)$ and a decoder $x^v = \mu(z)$. The latent T2V code $z \in \mathbb{R}^{h_z \times w_z \times c_z}$ is considered as generative feature.

CSPDarknet is employed as the backbone $\mathcal{F}$ to extract multi-level features c^3, c^4 and c^5 from the concatenated images x^t and x^v as follows,

$$[c^3, c^4, c^5] = \mathcal{F}(\text{concat}(x^t, x^v)). \quad (2)$$

A VMamba [35] block $\mathcal{M}$ then refines these features:

$$f^i = \mathcal{M}^i(c^i), \quad (3)$$

for $i \in \{3, 4, 5\}$. The output features $f^i \in \mathbb{R}^{h \times w \times c}$ are considered as discriminative features. We refer to this process as the VMamba-based Multi-level feature Enhancement (MEM). Here, a VMamba block is utilized to learn global relationships and contextual information from two modalities, thereby enhancing the model's representational capacity. Additionally, due to Mamba's input-dependent mechanism and various hardware-optimized techniques, it excels in long-range modeling with reduced computational cost compared to Transformers.

To better integrate the generated visible features, we propose a novel Graph-Mamba Fusion (GMF) module $\mathcal{G}$. This approach effectively combines generative feature z with discriminative features f_i as follows,

$$[g^3, g^4, g^5] = \mathcal{G}(f^3, f^4, f^5, z). \quad (4)$$

Graph-Mamba Fusion (GMF) module, as a key contribution, goes beyond simple attention by establishing high-level relationships. It first fuses generative feature z with the discriminate feature f^3 and then performs high-level fusion from f^3 to f^5 , which will be introduced later.

Subsequently, the features f^i and g^i are combined using residual connections and a Neck module Φ to obtain the output features $f^{i'} = \Phi(f^i + g^i)$, where $i \in \{3, 4, 5\}$. Finally, the Decoupled Detection Head $\mathcal{H}$ [12] is employed to obtain the detection result based on $f^{i'}$. The model is trained using the standard detection loss $L_d(\mathcal{H}(f^{3'}, f^{4'}, f^{5'}), y_{s_i}^t)$.

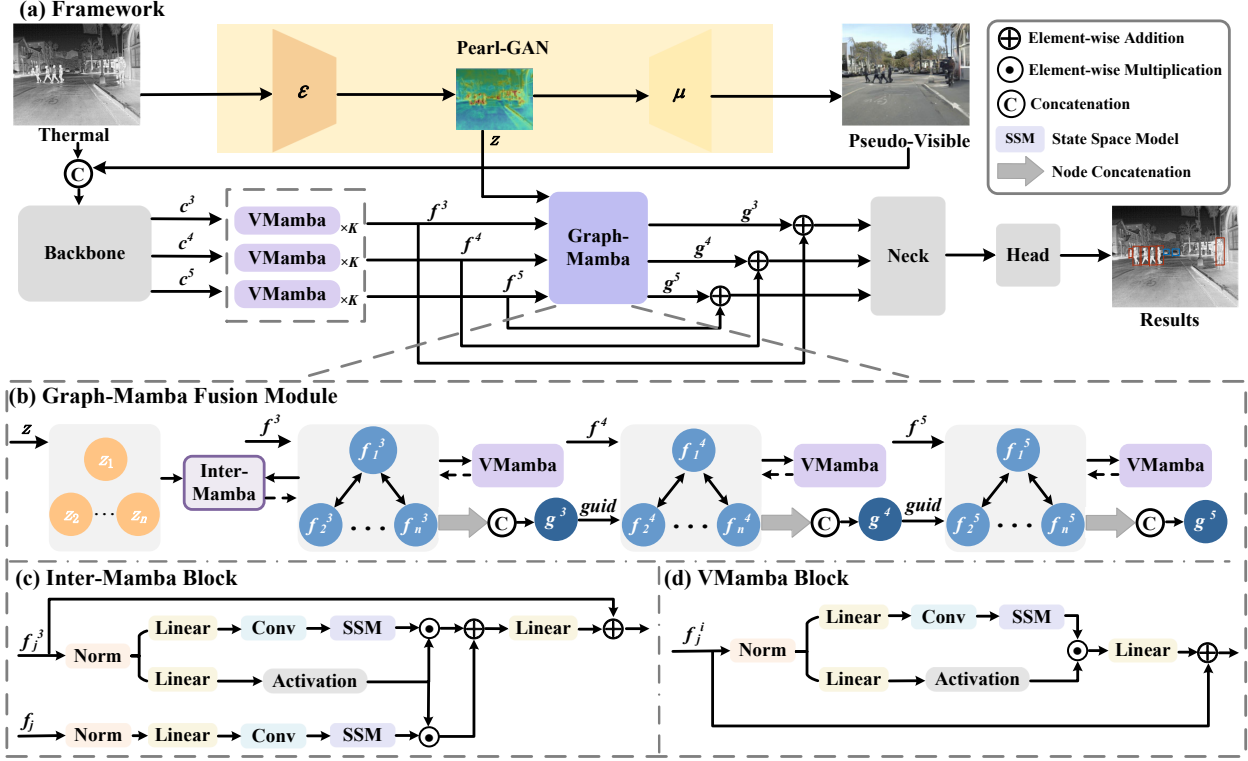


Figure 2. Overview of the proposed PFGF method. (a) The framework of the PFGF, with our modules highlighted in the purple box. (b) The Graph-Mamba Fusion Module constructs the graph structure for both discriminative and generative features, with leader nodes guiding the flow of information between graphs. (c) The Inter-Mamba Module facilitates interactions between different modalities. (d) The VMamba Module updates nodes within the same modality.

3.2. Pseudo Visible Feature Fine-Grained Fusion

For further integrating the generative feature z , a straightforward attention-based fusion approach, as used in [44], can be employed. However, this method has limitations in effectively capturing high-level relationships among the features. Additionally, it struggles to capture essential visual information due to the unreliability of the generated visible images, which often introduces a substantial amount of irrelevant information. To overcome these challenges, we introduce the Graph-Mamba Fusion (GMF) module as shown in Fig. 2(b). Graph structure is employed to establish more compensated relationships between different level features, which Mamba is adapted to capture long-range dependencies with less computational complexity.

To avoid getting lost in details, let's first briefly introduce the process. (1) Graph construction. The features f^3, f^4, f^5 , and z are down-and-up sampled to different copies as graph nodes. Graph edges are constructed by computing similarity between nodes, and message passing is performed by aggregating features from neighboring nodes. (2) Graph computation. An Inter-Mamba block is proposed to facilitate fusion between the nodes produced by latent feature z and discriminative feature f^3 respectively as illus-

trated in Fig. 2 (c). The GRU algorithm is used to update graph nodes, ensuring effective information propagation. Subsequently, a VMamba block is used to further refine the node feature. The output g^3 is formed by merging the node features of f^3 and is then multiplied with the nodes derived from f^4 , enabling the fusion of information across different levels. The output g^4 and g^5 are obtained by merging the node features from f^4 and f^5 , respectively.

3.2.1. Graph construction

In this part, we will introduce how to construct a graph $\Psi = (U, E)$ based on the features f and z . This structure is inspired by the method [24], which effectively models the relationships and dependencies among features extracted from different levels or modalities. The graph structure is designed to ensure that the fusion of these features is contextually relevant, thereby enhancing the accuracy and robustness of the representations used for detection.

Node generation. Given the generative feature z and discriminative feature f , two types of nodes are generated. To capture multi-level information and enhance feature diversity, we make replicas of different granularities of the corresponding features as graph nodes. Specifically, pyramid pooling, convolution, and upsampling operations are ap-

plied to both generative feature z and discriminate feature f , which can be formally computed as follows:

$$f_j^i = Up(Conv(\mathcal{P}_j(f^i))), z_j = Up(Conv(\mathcal{P}_j(z))), \quad (5)$$

where the operator $\mathcal{P}_j(\cdot)$ pools the given feature to various sizes according to the scale setting j , a 1×1 convolution is applied to reduce the dimension of the pooled representation, and Up operator, using bi-linear interpolation, up-samples the feature maps to match the original size. For each f^i and z , there are n node features $f_j^i \in R^{h \times w \times c'}$ for $j \in \{1, \dots, n\}$ and $z_j \in R^{h \times w \times c'}$. Here $n = 3$. Note that z has only one scale and shares the same shape as f^3 .

Edge generation. The edges play a crucial aspect, as it facilitates the propagation of information between nodes. Here, we only compute intra-edges, i.e., the edges connect the nodes produced by the same feature. Since our goal is to incorporate the information from the generative feature z into the discriminative feature f , there is no need to construct additional edges between the nodes produced by the generative feature. Specifically, for the feature f^i , the intra-edge $e_{j,k}^i$ is computed as

$$e_{j,k}^i = Conv(f_j^i - f_k^i), \quad (6)$$

where $Conv(\cdot)$ is used to learn relationships between nodes. The term $f_j^i - f_k^i$ represents the difference between the node features f_j^i and f_k^i . This highlights the relative disparity between nodes, following the approach outlined in [43]. Based on our edge computation method, we have formed four subgraphs, which are constructed by graph nodes formed by the features z, f^3, f^4 , and f^5 , respectively. Here the edges between the subgraphs of different discriminative features are not computed to avoid tedious computation. Instead, we introduce a new leader node to aggregate subgraph node features, and then pass the aggregated feature to the next subgraph from f^3 to f^5 .

Message Passing. In each subgraph composed of discriminative features, every node aggregates feature messages from all its neighboring nodes. For the node f_j^i , at $l-1$ message passing, denoted as $m_j^{i,l-1}$, is computed as

$$m_j^{i,l-1} = \sum_{k \in \mathcal{N}(j)} Sigmoid(e_{k,j}^{i,l-1}) \cdot f_k^{i,l-1}, \quad (7)$$

where $Sigmoid(\cdot)$ function is utilized to calculate the link weight between nodes.

3.2.2. Graph computation

Graph-based cross-modality information fusion consists of two key components: cross-modality fusion through the Inter-Mamba block and knowledge integration via GRU and VMamba.

Cross modality fusion by Inter-Mamba block. For fusing the node features z_j and f_j^3 , instead of transformer-based

route, we proposed a Mamba-based block. As shown in Fig. 2(c), the node features f_j^3 and z_j are first projected into the hidden state space followed by a VMamba block, and a gating operation is then employed to facilitate the interaction between these two types of hidden state features.

Specifically, the node features f_j^3 and z_j are first flattened to form the input token sequences T_j^3 and T_j , each with dimensions $B \times N \times C$, where B , N , and C denote the batch size, sequence length, and number of channels, respectively. These sequences are then normalized by a normalization layer. Next, the normalized sequences T_j^3 and T_j are projected into the hidden state space respectively through linear layers, convolution, and SSM, resulting in the hidden state features $\hat{f}_j^3$ and $\hat{z}_j$. Simultaneously, the normalized sequence T_j^3 is also projected through linear layers with SiLU activation to obtain the gating weight r . It is used for hidden state feature fusion as follows,

$$f_j^3 = \mathcal{R}(\mathcal{Y}((\hat{z}_j + \hat{f}_j^3) \odot r) + f_j^3), \quad (8)$$

where $\odot$ represents element-wise multiplication. The operator $\mathcal{Y}$ denotes the projection operation with a linear transformation, while $\mathcal{R}$ refers to the reshaping operation.

Cascade Knowledge Integration (CKI) strategy. Existing studies [11] have demonstrated that deep and shallow features exhibit complementary characteristics: deep features capture high-level semantic information, while shallow features retain rich spatial details. Leveraging multi-level features is therefore advantageous for enhancing detection performance. As we stated before, a leader node is proposed to convey low-level fused information to high-level subgraph in a cascade manner.

We first apply a GRU-based update method [9] to enhance the nodes' ability to capture complex feature interactions and retain relevant information from neighboring nodes. After $l-1$ message passing, the node feature f_j^i is updated by $\mathcal{U}_{GRU}(f_j^i, m_j^{i,l-1})$ (See the Appendix for a detailed description of $\mathcal{U}_{GRU}$).

A leader node feature g^i at the i -th scale is computed as

$$g^i = Conv(Concat(\mathcal{M}(f_1^i), \dots, \mathcal{M}(f_n^i))), \quad (9)$$

where $\mathcal{M}$ denotes a VMamba block as mentioned before.

The leader node g^i conveys knowledge to the subsequent subgraph nodes as follows,

$$f_j^{i+1} = f_j^{i+1} \odot \mathcal{A}(g^i) + f_j^{i+1}, \quad (10)$$

where $\mathcal{A}$ represents an attention operation, which is computed via global average pooling followed by a Sigmoid activation. Since the last subgraph does not contribute to knowledge transmission, the scale i is set to 3 and 4.

4. Experiments

4.1. Experimental Settings

Datasets. To evaluate our method, we conduct experiments on the FLIR-aligned dataset [49], LLVIP dataset [18], and Autonomous Vehicles dataset [42]. (1) The FLIR dataset originally presented challenges due to unaligned thermal-visible image pairs. Therefore, we utilize the aligned version, consisting of 5,142 thermal-visible image pairs, with 4,129 pairs for training and 1,013 for testing. This dataset includes three object categories: “person,” “car,” and “bicycle”. (2) The LLVIP dataset is collected in very dark environments. It comprises 15488 aligned image pairs, with 12,025 pairs used for training and 3,463 pairs for testing. This dataset focuses on a single object class: “person”. (3) The Autonomous Vehicles dataset comprises 6,009 paired images for training and 1,503 for testing, covering five categories: “bike”, “car”, “car_stop”, “color_cone”, and “person”. Compared to the FLIR and LLVIP datasets, the scenes in this dataset are more complex.

Evaluation Metrics. We evaluate our method using standard object detection metrics [31], including mean Average Precision (mAP), mAP50, and mAP75. The mAP metric represents the average AP over IoU thresholds ranging from 0.50 to 0.95, with an interval of 0.05. The mAP50 and mAP75 metrics correspond to the AP values at IoU thresholds of 0.50 and 0.75, respectively. Higher values for these metrics indicate better performance.

Implementation Details. We utilize a pre-trained model on the COCO dataset and implement our approach using the MMDetection toolbox [6], a widely adopted framework for object detection. The network is trained using the SGD optimizer with a batch size of 4 and an initial learning rate of $5e-4$. The input image size is set to 640×640 for the FLIR, LLVIP, and Autonomous Vehicles datasets, with the training conducted over 8 epochs. The training process is conducted in two stages. In the first stage, the Pearl-GAN module is trained. In the second stage, with the parameters of the Pearl-GAN module fixed, the backbone, VMamba, Graph-Mamba Fusion (GMF) module, neck, and detection head are trained. For the GMF module, the number of graph nodes n is set to 3, and the number of message-passing iterations L is also set to 3. The cascade quantities of VMamba and Inter-Mamba blocks are each configured to 2. Details on hyper-parameter sensitivity are provided in the Appendix. To ensure reproducibility, all experiments are conducted with a random seed set to 0.

4.2. Comparison with State-of-the-art Methods

Compared methods. We compare the proposed PFGF method with both mono-modality (thermal) and multi-spectral (visible + thermal) methods. The mono-modality methods include Faster RCNN [41], SSD [33], RetinaNet

Methods	Data	Backbone	mAP50	mAP75	mAP
Faster RCNN [41]	T	ResNet50	74.4	32.5	37.6
SSD [33]	T	VGG16	65.5	22.4	29.6
RetinaNet [32]	T	ResNet50	64.5	20.3	28.3
YOLOv3 [40]	T	Darknet53	73.6	31.3	36.8
YOLOX-L [12]	T	CSPD53	78.1	39.4	42.6
IAH [45]	T	Darknet53	76.4	-	-
DIP [17]	T	ResNet50	77.3	-	-
TIRDet [44]	T	CSPD53	81.4	41.1	44.3
IDA [28]	T	CSPD53	81.1	43.4	46.3
PFGF (Ours)	T	CSPD53	84.8	44.6	47.1
CFR_3 [49]	V+T	VGG16	72.4	-	-
GAFF [50]	V+T	ResNet18	72.9	32.9	37.5
YOLOFusion [39]	V+T	VGG16	76.6	-	39.8
CMX [51]	V+T	SwinT	82.2	37.1	42.3
IGT [7]	V+T	SwinT	85.0	36.9	43.6
DATFF [16]	V+T	ResNet50	80.2	-	-
RSDet [52]	V+T	ResNet50	81.1	-	41.1
Fusion-Mamba [10]	V+T	CSPD53	84.9	-	47.0

Table 1. Comparisons on FLIR dataset in terms of MAP. The best results are highlighted in bold.

[32], YOLOv3 [40], YOLOX-L [12], IAH [45], DIP [17], TIRDet [44] and IDA [28]. These methods utilize paired visible and thermal images for training, but only thermal images for testing. In contrast, the multispectral methods include CFR_3 [49], GAFF [50], YOLOFusion [39], CMX [51], IGT [7], DATFF [16], RSDet [52], YOLOv5-VT [19], DM-Fusion [47], and Fusion-Mamba [10]. These methods employ paired visible and thermal images for both training and testing. The results presented in the tables are cited from the respective papers.

Quantitative comparison. The quantitative comparison results are presented in Tables 1, 2, and 3. Our method achieves the state-of-the-art (SOTA) performance among mono-modality (thermal) methods on the FLIR, LLVIP, and Autonomous Vehicles datasets. On the FLIR dataset, our approach outperforms the recent TIRDet and IDA methods, achieving mAP gains of 2.8% and 0.8%, respectively, along with mAP50 improvements of 3.4% and 3.7%. For a fair comparison of the LLVIP dataset, we reproduced the TIRDet method using a Pearl-GAN model trained on the LLVIP dataset, referred to as TIRDet*. Similarly, our method denoted as PFGF*, utilizes this same Pearl-GAN model. Additionally, to facilitate comparison with the original results reported in the TIRDet paper, we also use a Pearl-GAN model trained on the FLIR dataset. The results demonstrate that our proposed method consistently achieves superior performance in both sets of comparisons: between TIRDet and PFGF, as well as between TIRDet* and PFGF*. On the more complex Autonomous Vehicles dataset, our method surpasses the IDA method, with mAP50 increas-

Methods	Data	Backbone	mAP50	mAP75	mAP
Faster RCNN [41]	T	ResNet50	96.1	68.5	61.1
SSD [33]	T	VGG16	90.2	57.9	53.5
RetinaNet [32]	T	ResNet50	93.7	49.3	50.9
YOLOv3 [40]	T	Darknet53	89.7	53.4	52.8
YOLOX-L [12]	T	CSPD53	95.2	72.7	63.6
TIRDet [44]	T	CSPD53	96.3	73.1	64.2
TIRDet* [44]	T	CSPD53	96.8	74.7	65.2
PFGF (Ours)	T	CSPD53	96.6	77.9	66.8
PFGF* (Ours)	T	CSPD53	96.9	78.4	67.3

DM-Fusion [47]	V+T	CSPD53	88.1	-	53.1
YOLOv5-VT [19]	V+T	CSPD53	95.8	71.4	62.3
RSDet [52]	V+T	ResNet50	95.8	-	61.3
Fusion-Mamba [10]	V+T	CSPD53	97.0	-	64.3

Table 2. Comparisons on LLVIP dataset in terms of MAP.

Methods	Data	Backbone	mAP50	mAP75	mAP
YOLOv3 [40]	T	Darknet53	71.9	29.9	34.6
FasterRCNN [41]	T	ResNet50	72.5	39.8	39.9
CascadeRCNN [4]	T	ResNet50	72.9	42.7	42.2
YOLOX-L [12]	T	CSPD53	72.3	40.6	40.6
IDA [28]	T	CSPD53	73.1	43.8	43.1
PFGF (Ours)	T	CSPD53	78.8	46.6	45.9

Table 3. Comparisons on AV dataset in terms of MAP. AV: Autonomous Vehicles dataset.

ing by 5.7% , mAP75 rising by 2.8%, and overall mAP improving by 2.8%. Moreover, our method outperforms most multispectral methods in detection performance, achieving the highest mAP and mAP75, and the second-best mAP50 on both FLIR and LLVIP datasets. These results indicate that the combination of the T2V model and the proposed PFGF approach effectively enhances thermal detection performance with valuable visible information.

We also conduct a visual comparison between the proposed PFGF method, and the publicly method TIRDet. Our method significantly improves the detection performance of thermal images (See the Appendix).

4.3. Further Analysis

Ablation studies. Our framework comprises T2V translation model Pearl-GAN, VMamba-based Multi-level Feature Enhancement (MEM) block, and Graph-Mamba Fusion (GMF) module, which includes both the Inter-Mamba block and CKI strategy. To assess the contribution of each component, we performed ablation studies on the FLIR, LLVIP, and Autonomous Vehicles datasets. There are several variants: ‘w/o Pearl-GAN’ refers to the exclusion of visible information with only thermal image input; ‘w/o MEM’ denotes the detector backbone features are not enhanced by MEM block; ‘w/o GMF’ means the enhanced

Method	mAP		
	FLIR	LLVIP	AV
w/o PearlGAN	45.2	65.0	44.4
w/o MEM	46.0	66.1	45.4
w/o GMF	45.6	65.6	44.3
w/o Inter-Mamba (GMF)	45.9	66.2	45.1
w/o CKI (GMF)	46.1	66.4	45.3
PFGF	47.1	67.3	45.9

Table 4. Ablation study on FLIR, LLVIP, and AV datasets.

features will be passed directly to the detector head; ‘w/o Inter-Mamba’ implies that generative features z are not fused with discriminative features f^3 ; while ‘w/o CKI’ does not integrate features from f^3 to f^5 .

Table 4 demonstrates that each component positively contributes to performance. Removing Pearl-GAN prevents the incorporation of visible information, leading to a substantial performance decline on the FLIR and LLVIP datasets and the second-largest performance drop on the Autonomous Vehicles dataset. The absence of MEM affects performance differently across datasets, suggesting its role in refining and enhancing coarse fusion features. Excluding the GMF module reduces mAP by 1.5% on the FLIR dataset, 1.7% on the LLVIP dataset, and 1.6% on the Autonomous Vehicles dataset, underscoring its importance in integrating generated visible and thermal information. Furthermore, ablation analysis on the GMF module reveals that the Inter-Mamba block has a more significant impact than the CKI block, indicating that the Inter-Mamba block can effectively filter redundant visible information while preserving information beneficial for thermal detection.

Comparison of different fusion methods. To validate the superiority of our proposed fusion module, we conduct further experiments comparing it with various fusion strategies, including addition, concatenation, CMA [44], MMFT [45], GIM [24] and FMB [10]. The CMA method employs Convolutional Block Attention Module (CBAM) to generate attention maps; the MMFT leverages a Transformer-based architecture to perform fusion based on varying illumination conditions; GIM utilizes graph neural networks to handle cross-modal interactions through features; while the FMB method, rooted in the Mamba architecture, is specifically designed for multispectral detection tasks.

As shown in Table 5, our method outperforms all other strategies. Specifically, CMA method is relatively simplistic and fails to effectively capture visible information beneficial for thermal detection; in contrast to the MMFT method, our approach eliminates the dependency on an additional illumination-aware module. The FMB method has achieved superior performance in multispectral tasks, it does not adapt well to our specific problem due to the unreliability of the generated visible information. The GIM

Method	FLIR		LLVIP		AV	
	mAP	mAP50	mAP	mAP50	mAP	mAP50
Add	42.6	77.5	61.7	95.2	40.4	73.9
Concat	43.7	79.9	63.3	94.8	42.4	74.7
CMA[44]	44.3	81.4	64.6	96.3	43.9	76.5
MMFT[45]	44.6	82.1	65.1	95.2	44.1	76.8
GIM[24]	46.0	81.8	66.3	96.2	44.8	74.5
FMB[10]	45.7	82.4	65.9	96.3	44.2	77.0
GMF (Ours)	47.1	84.8	67.3	96.9	45.9	78.8

Table 5. Comparison of different fusion methods.

Method	mAP	mAP50	FLOPS(G)↓	Param(M)↓
GMTF	46.2	82.7	604.1	71.8
PFGF	47.1	84.8	409.0	66.9

Table 6. Comparisons of VMamba and Transformer based methods on the FLIR dataset.

method introduces redundant information when facilitating interactions among features at various granularities. While our GMF method outperforms the FMB fusion method which depends exclusively on Mamba. These results indicate that the Graph-Mamba-based architecture facilitates a more explicit and robust integration of thermal and pseudo-visible information.

The effectiveness of using VMamba block. An efficiency analysis is conducted to evaluate Mamba block compared with Transformer block. For a fair comparison, all VMamba blocks are replaced with Transformer blocks, resulting in a new method designated as GMTF. Specifically, the VMamba block is substituted with a self-attention block, and the Inter-Mamba block is replaced with a cross-attention block. As shown in Table 6, our PFGF method surpasses GMTF with a 0.9% increase in mAP and a 2.1% gain in mAP50. Additionally, PFGF demonstrates parameter efficiency by reducing the parameter count by 4.93M and decreasing FLOPs from 604.1 GFLOPs to 409.0 GFLOPs. These findings highlight that the VMamba block not only effectively suppresses redundant features and captures complementary cross-modal information but also achieves superior computational efficiency.

The effectiveness of using leader nodes. To incorporate the generative feature z into the multi-level discriminative features, we utilize leader nodes to guide information transfer from the low-level subgraph to the high-level subgraph. To verify the effectiveness of leader nodes, we compare our method with an approach that lacks leader nodes on FLIR and LLVIP datasets. Without leader nodes, the next high-level features don’t receive the previous low-level features, which ignore the integration from f^3 to f^5 . With ground truth serving as a labeled reference, Fig. 3 clearly shows that without the assistance of leader nodes, the object

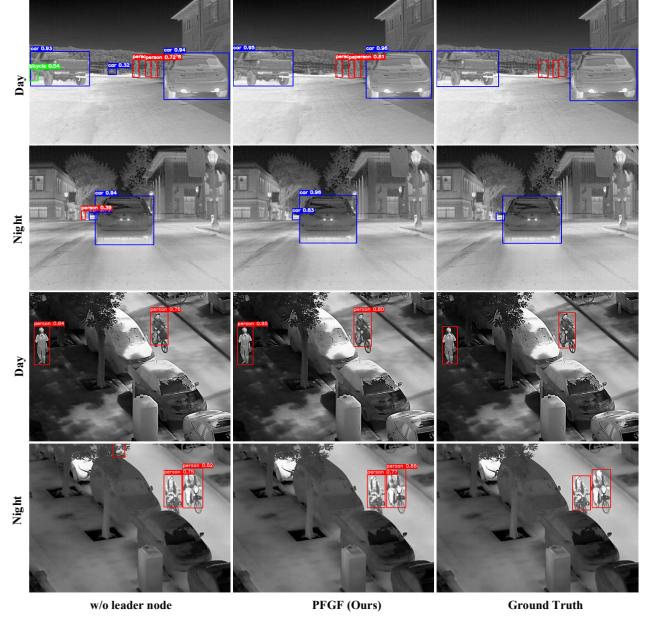


Figure 3. Effectiveness of using leader nodes. The first two rows present samples from FLIR dataset, and the last two rows present samples from LLVIP dataset. Red, blue, and green boxes denote the detected objects “person”, “car”, and “bicycle”, respectively.

confidence is significantly lower and there are more occurrences of false detections. This indicates that our method effectively leverages the features delivered by leader nodes across subgraphs, integrating both high-level semantics and low-level local details to build a more robust representation.

5. Conclusion

In this paper, we proposed a novel cross-modal fusion method, termed Pseudo Visible Feature Fine-Grained Fusion (PFGF), for thermal object detection. The PFGF framework begins by constructing a graph which consists of several subgraphs. Each subgraph is generated from different level output features from the backbone or synthetic visible feature. An Inter-Mamba block is then employed to facilitate cross-modality feature association between generative and discriminative node features. Additionally, we introduce a Cascade Knowledge Integration strategy to propagate knowledge hierarchically from lower-level to higher-level subgraphs. In the process, VMamba is employed to further refine features. Extensive experiments demonstrate that our method achieves the SOTA performance.

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